



**EN-MADISYN-LE-154-94-0705 (H-7, H-8, H-9, H-10, H-11, H-12)
Spud Meeting**

Isabel Leon

08/04/2023

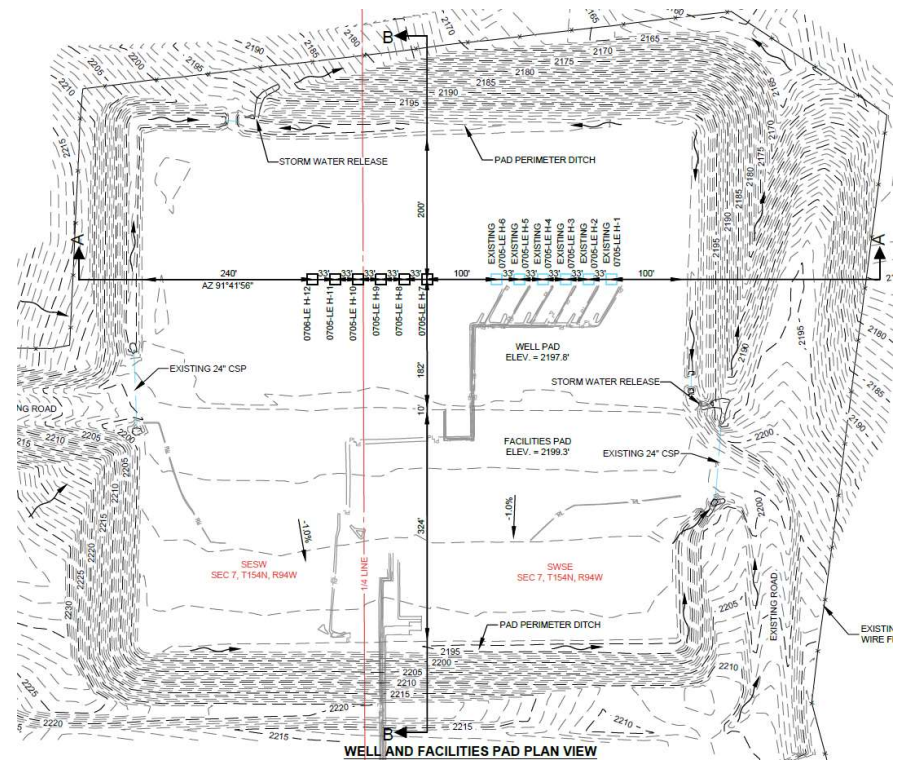
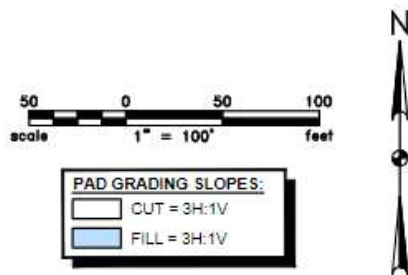
EN-MADISYN-LE-154-94-0705 (H-7, H-8, H-9, H-10, H-11, H-12)

Drilling Order



3F	EN-MADISYN-LE-154-94-0705H-7	10.3 days	8/13/2023
MB	EN-MADISYN-LE-154-94-0705H-8	10.3 days	8/23/2023
3F	EN-MADISYN-LE-154-94-0705H-9	10.3 days	9/3/2023
MB	EN-MADISYN-LE-154-94-0705H-10	10.3 days	9/14/2023
3F	EN-MADISYN-LE-154-94-0705H-11	10.2 days	9/25/2023
MB	EN-MADISYN-LE-154-94-0706H-12	10.2 days	10/6/2023

- H-7 → H-8 → H-9 → H-10 → H-11 → H-12

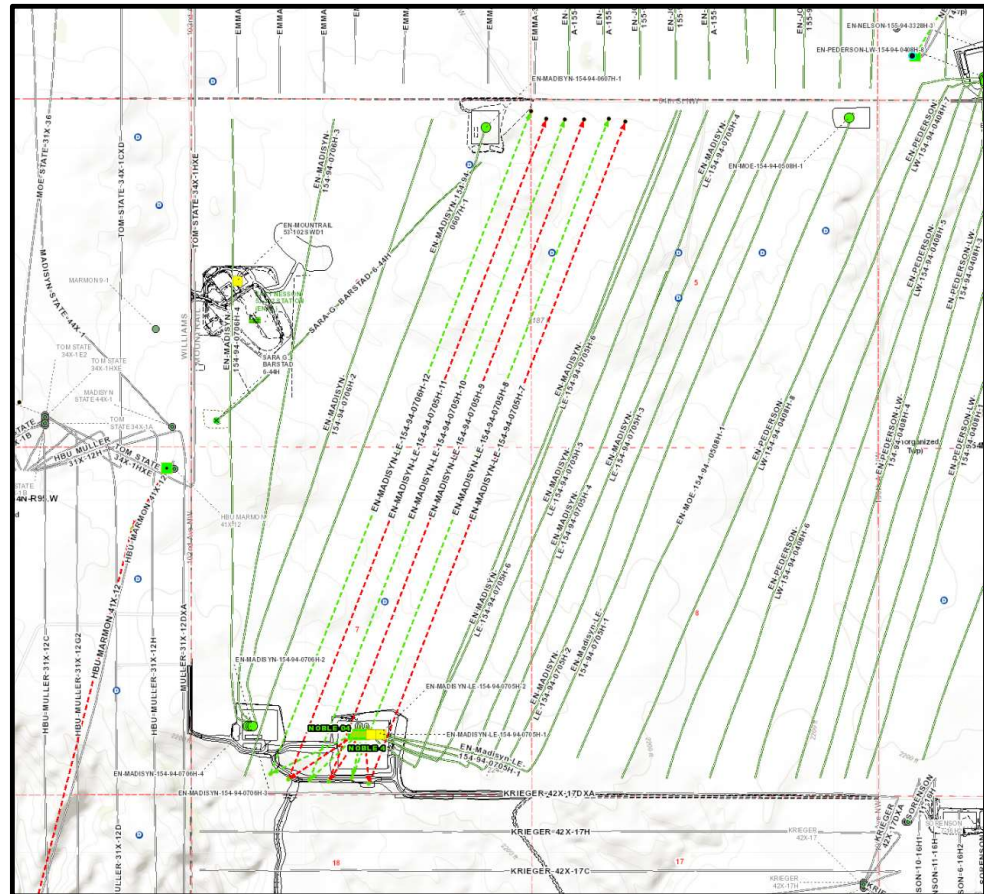


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Offset Hazards

- Offsets
 - No operational issues on offset wells
- SWD:
 - 5 wells identified
- AC concerns
 - None

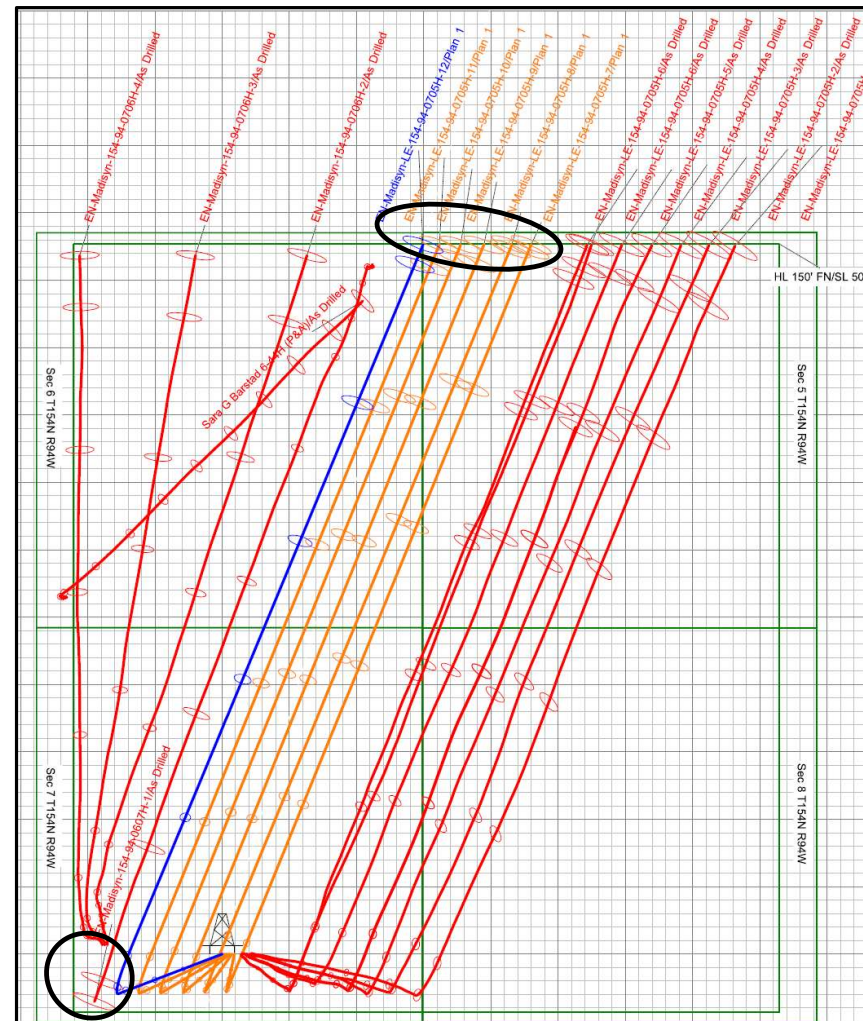


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SIMOPS & AC Concerns

- Do not cross section line buffer:
 - Keep communication with RTC to plan for any needed extra footage.
- Offset well EN-Madisyne H-1



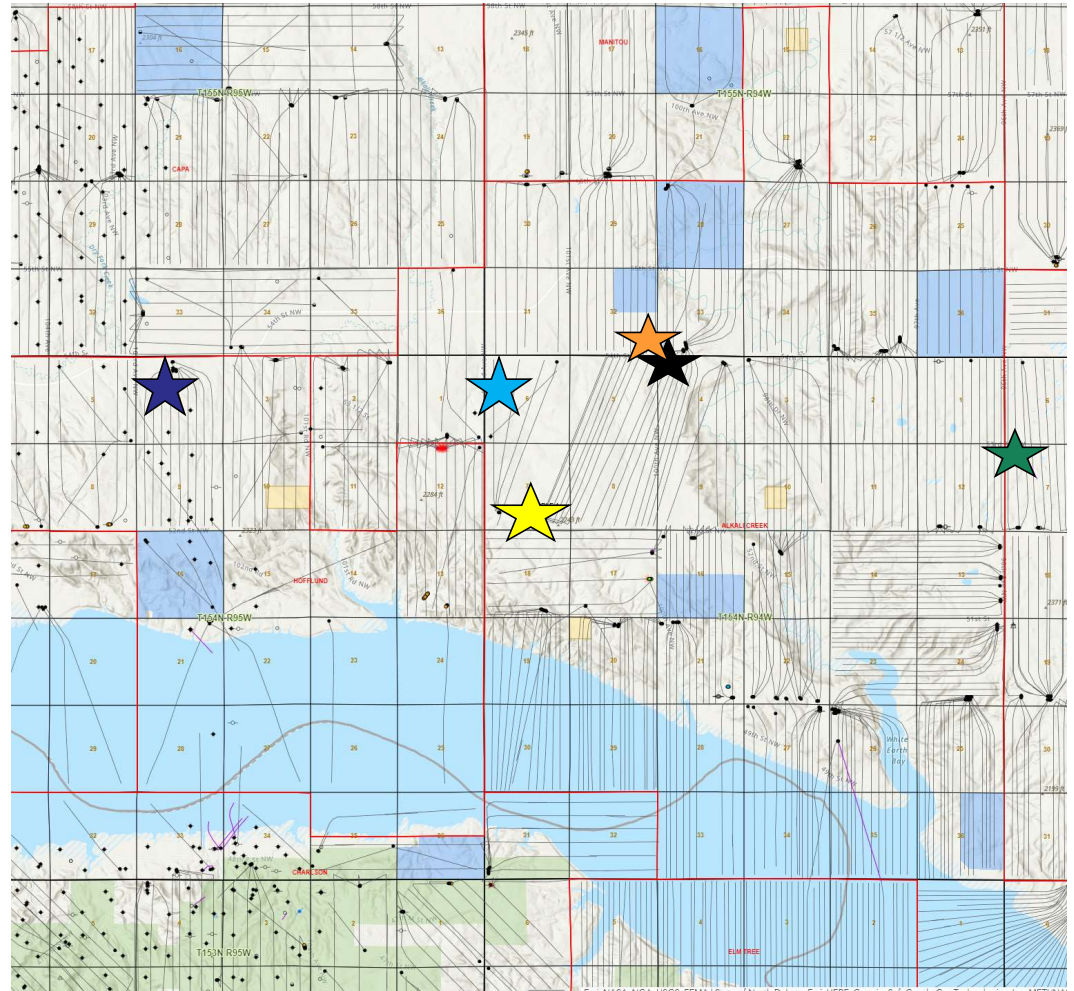
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Low Risk



	SWD Well	Operator	Distance	Inj Rate (bwpd)	Flowing Tbg Pressure (psi)
★	NELSON SWD 2	Tallgrass	2.5 mi NE of pad	10,525	1,005
★	NELSON SWD 14-33H	Tallgrass	2.4 mi NE of pad	3,642	582
★	EN-MOUNTAIL 53-102 SWD1	Hess	1.3 mi N of pad	6,279	374
★	HOFFLUND 3	Rim Operating	4.0 mi NW of pad	335	447
★	BILLADEAU SWD 5493 11-7	CrestWood	5.7 mi E of pad	2,493	445

★ Pad Location



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Drilling Outline



EN-MADISYN-LE-154-94-0705H-7							
Supplement / Drilling Outline							
Rig: NaborsX10 API No.: 33-06-05177 Oil Field: ALKALI CREEK WBS / Network # 105-05027.04.01/ Permit/WF#: 39891 Objective: 3F FirstBench KB Height, ft: 32.0		Co., State: MOUNTRAIL, ND Surf Location: Sec. 7 154 N 94 W BH Location: 160 ft FNL & 1490 ft FWL, SECTION 05, T154N, R94W Hard Line: 150°FNL/100°FSL Latitude: 48° 10' 15.55" Longitude: -102° 49' 2.45" GL Elevation: 2,198 ft, graded, above MSL		Notes & Deviation: Three Forks Lateral TD well at least 160ft from North Section Line			
Formation Tops TVD	Planned Wellpath	Hole Size/Depth	Mud Properties	Offset Hazards and Risks	Proposed BHA	Casing / Completions Program	Cement Program
Pierre 1,835 Greenhorn 4,288 Mowry 4,666 Swift 5,460 Rierdon 5,887 Piper Lime 6,184 Top_Dunham_Salt 6,424 Speartfish 6,471 Top_Pine_Salt 6,769 Top_Opeche_Salt 7,001 Amsden 7,378 Tyler 7,561 Kibbey Lime 8,145 Charles 8,315 Ratcliffe 8,933 Top_Last_Salt 9,017 Base_Last_Salt 9,966 Rival 9,203 Lodgepole 9,857 Bakken 10,603 Middle_Bakken 10,625 Base_Mid_Bakken 10,687 Three_Forks 10,743	KB, ft 32 3.8 Tangent, deg 10,201 TOL, MD 10,301 KOP, MD 12.5 deg/100 ft Build	16 in. Surf to 80 ft MD/TVD 13-1/2 in. Surf to 1,885 ft MD 8-3/4 in. 1,885 ft MD/TVD to 11,053 ft MD 10,772 ft TVD 6 in. 11,053 ft MD to 21,365 ft MD 10,708 ft TVD	8-3/4" Intermediate Section: Halliburton Integrate Bara-Xcel OBM Shoe to KOP: 9.5 - 9.8 ppg TD KOP w/ 9.5 - 9.8 ppg Enter UBS w/ 10.0 - 10.2 ppg Land Curve w/ 10.2 ppg HTHP Filtrate: < 4ml OWR: 60/40 Tau 0: 3 - 6 Utilize centrifuge as needed to keep LGS below 7.5%. 6" Lateral Section: Salt Water MW: 9.0-10.0 ppg LGS: < 5% pH: 8 - 9 Pump sweeps as needed using Drillchem HE Lube	- One active offset SWD well 1.3 miles away (EN-MOUNTRAIL 53-102 SWD) operated by Hess). - One active offset SWD well 2.5 miles away (NELSON 14-33H operated by Tallgrass Water North Dakota). - One new offset SWD well 2.4 miles away (NELSON SWD 2 operated by Tallgrass Water North Dakota). - One active offset SWD well 4.0 miles away (Hofflund 3 operated by Rim Operating). - One active offset SWD well 5.7 miles away (Billdeau SWD 5493 11-7 operated by CrestWood). 6" Lateral Section: MW: 9.0-10.0 ppg LGS: < 5% pH: 8 - 9 Pump sweeps as needed using Drillchem HE Lube	8-3/4" Vertical BHA (1st BHA): 8-3/4" TKC66 BP1 or RPS 616 7-1/4" ERT NOV 1.5", 0.23 RPG, 7/8, 5.7 Stg; 1,185 psi max Differential P. at 180 F MWD Gordon Filter Sub 5" NC50 HWDP (21 js) 5" NC50 DP 8-3/4" Vertical BHA (2nd BHA): 8-3/4" NOV TKC66 BC2 bit 7-1/4" ERT NOV 1.5", 0.23 RPG, 7/8, 5.7 Stg; 1,185 psi max Differential P. At 180 F MWD Gordon Filter Sub 5" NC50 HWDP (21 js) 5" NC50 DP 8-3/4" Curve BHA: 8-3/4" ULTERRA CF516 Single row bit 6-3/4" ERT NOV 2.38", 0.47 RPG, 4/5, 7.0 Stg; 1,290 psi max Differential P. NMDC/MWD Gordon Filter Sub 5" NC50 HWDP (21 js) 5" NC50 DP 6" Lateral BHA: 6" Security GT64DK or NOV TKC63 X6 bit 5.25" NOV 1.5", 0.60 rpg, 6/7, 7.8 Stg MM NMDC/MWD Gordon X/O Filter Sub 4" DP (+/- 120 js) 5" Agitator/XRV	9-5/8 in. Surface 9-5/8 in. 36 lb/ft J-55 STC Preset at 1,989 ft ref GL Note: Got back 61 bbl of cement returns to surface. 7 in. Production 7 in. 32 lb/ft P-110RY BTC Surface to 6,300 ft 7 in. 32 lb/ft P-110EC BTC 6,300 ft to TD 4-1/2 in. Liner 13.5 lb/ft P110 DWCC - Curve 11.6 lb/ft P110 BT&C - Lateral 4-1/2 in. BTC Frac String	9-5/8 in. Cement Lead 60 % Exces Type 1L 12.0 ppg 2.52 cu ft/sack Tail: 20 % Exces Type 1L 13.0 ppg 1.98 cu ft/sack Cement Service Provider: Halliburton (13/14.7Elasticem) 7 in. Cement Spacer: 50 bbl Gelled Salt Water 60 bbl Tuned Spacer Top of Lead Cement, ft 1,885 13.0 ppg 1.94 cu ft/sack 344 sks Type 1L Top of Tail Cement, ft: 5,924 14.7 ppg 1.95 cu ft/sack 450 sks Type 1L 409 bbl Approx. Displacement 4-1/2 in. Cement 14.3 ppg 1.56 cu ft/sack 685 sks Class G
22 deg Azimuth to land 7 in. 22 deg Lateral Drilling Azimuth Formation Dip ±90.36 S to N							
ORGANIZATION APPROVALS							
Tom Richardson Drilling Team Lead		Jim Stover Drilling Superintendent		Isabel Leon Drilling Engineer			

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File Locations



<https://teams.microsoft.com/l/channel/19%3a565c31e6aa9146458452e67f58bf8fc7%40thread.tacv2/EN-MADISYN-LE-154-94-0705H-7?groupId=02096943-4642-4ef9-9bab-21969d78a070&tenantId=8f12f317-2734-454e-8e5c-3941f8946c51>

The screenshot displays the Microsoft Teams interface for the 'EN-MADISYN-LE' channel. The channel name is circled in red. The 'General' tab is selected, showing a list of documents. The document 'EN-MADISYN-LE-154-94-0705H-7' is highlighted in red. The document list includes folders for 'Hazards', 'Pad Layout', and 'Permits'. The 'EN-MADISYN-LE-154-94-0705H-7' document is a folder containing various files, including 'BHA', 'Cement', 'Directional', 'Drilling Outline', 'Geology', and 'Mud'. The document is modified on July 27 by Leon, Isabel.

Name	Modified	Modified By
BHA	July 27	Leon, Isabel
Cement	July 27	Leon, Isabel
Directional	July 27	Leon, Isabel
Drilling Outline	July 27	Leon, Isabel
Geology	July 27	Leon, Isabel
Mud	July 27	Leon, Isabel